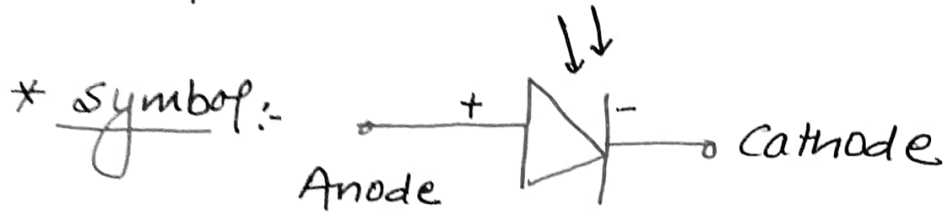


* It Converts light energy into electrical signals⁽¹⁾

PhotoDiode :- Photo diode is a special purpose diode

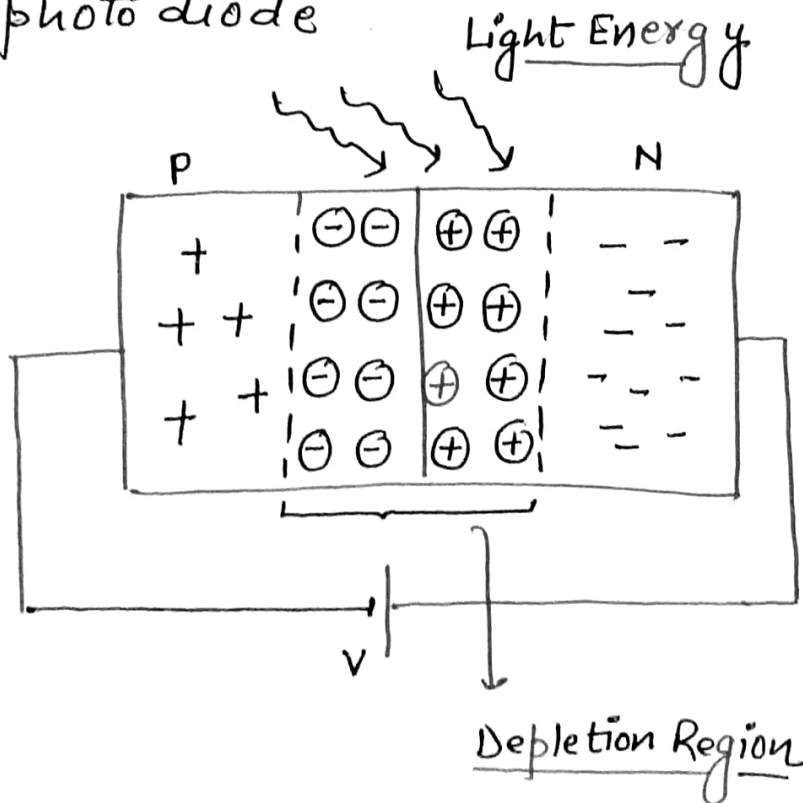
Which operates in the reverse biased condition.

* It is also known as photo detector, Light detector or photosensor.



Construction & Working:- The Construction of photo diode is similar to the

Construction of p-n junction diode except that the light energy is incident on depletion region in case of photo diode



②

Working:- When no light is incident on depletion region, photodiode behaves like a p-n junction diode in reverse biased condition where a small leakage current flows due to minority charge carriers. This current (in the absence of light) is called dark current.

⇒ In order to increase this dark current (μA), the light energy is incident on the depletion region.

⇒ When the energy of the ^{incident} light becomes greater than the forbidden energy gap, the bonds (im-mobile charge carriers) in the depletion region are broken and free electron-hole pairs are generated.

⇒ These free electrons and holes get attracted towards (+ve) plate and (-ve) plate of the battery respectively. Therefore substantial current flows through the photodiode. This current is known as photocurrent.

⇒ This photocurrent depends on the intensity of the incident light. (directly proportional).

Applications:-

- ① It is used in alarm system
- ② In Industrial automation
- ③ In Medical devices

① PIV (Peak Inverse Voltage) :- It is defined as a maximum negative voltage that a diode can withstand before getting damaged when it is operated in reverse biased condition. (3)

* PIV for Half Wave Rectifier $= V_m$ (peak value of A.C input voltage)

* PIV for Full Wave Rectifier :-

$$\left. \begin{array}{l} \text{For Centre tapped F.W.R} = 2V_m \\ \text{for Bridge F.W.R} = V_m \end{array} \right\}$$

② Ripple factor :- It indicates that how close the rectified output is to the pure D.C. voltage.

Ripple factor $\xi = \sqrt{\left(\frac{I_{rms}}{I_{dc}}\right)^2 - 1}$

* Its value should be as small as possible

$$\Rightarrow \left. \begin{array}{l} \text{Ripple factor for H.W.R} = 1.21 \\ \text{Ripple factor for F.W.R} = 0.482 \end{array} \right\}$$

Q) for a given transistor, Current Gain for CB (common Base) Configuration is 0.98, collector Current is 98 mA. find base Current.

Solution: Given Current Gain $\alpha = 0.98$

Collector Current $I_C = 98 \text{ mA}$

Base Current $I_B = ?$

We know $\alpha = \frac{I_C}{I_E}$

$$I_E = \frac{I_C}{\alpha} = \frac{98}{0.98} \Rightarrow \boxed{I_E = 100 \text{ mA}}$$

We know $I_E = I_B + I_C$

$$I_B = I_E - I_C$$

$$= 100 - 98$$

$$\boxed{I_B = 2 \text{ mA}}$$

$$\text{or } \boxed{I_B = 2000 \mu\text{A}}$$

$$\boxed{I_B = \cancel{0.002 \text{ A}}}$$

Ans